

ASSIGNMENT SUBMISSION

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1. Relational Databases (RDBMS)

- Store data in tables (rows and columns)
- Use a fixed schema (structure defined in advance)

Key Features

- Structured data format
- Use SQL (Structured Query Language)
- Tables linked using primary keys and foreign keys
- Support ACID properties (Atomicity, Consistency, Isolation, Durability)
- Ensure high data integrity and consistency

Advantages

- Reliable and accurate data handling
- Good for complex queries
- Strong relationships between data

Disadvantages

- Less flexible when changing structure
- Harder to scale horizontally

Examples

- MySQL
- PostgreSQL
- Oracle Database
- Microsoft SQL Server

2. Non-Relational Databases (NoSQL)

Definition

- Store data in non-tabular formats
- Use a flexible or dynamic schema

Key Features

- Can store:
 - Documents (JSON)
 - Key-value pairs
 - Graphs
 - Wide-column data
- No fixed schema
- Designed for large-scale and distributed systems
- Support horizontal scaling

Advantages

- Highly scalable
- Flexible data structure
- Good for unstructured or semi-structured data

Disadvantages

- Less strict data consistency
- No standard query language

Examples

- MongoDB
- Redis
- Cassandra
- Neo4j

3. Key Differences

Structure

- Relational: Tables (rows & columns)
- Non-relational: Flexible formats (JSON, key-value, graph)

Schema

- Relational: Fixed
- Non-relational: Dynamic

Query Language

- Relational: SQL
- Non-relational: No standard language

Scalability

- Relational: Vertical scaling
- Non-relational: Horizontal scaling

Data Consistency

- Relational: Strong consistency (ACID)
- Non-relational: Often eventual consistency

Best Use Cases

- Relational: Banking, accounting, inventory systems
- Non-relational: Social media, big data, real-time apps